

Presence-Gated Autonomy: A Systems-Leverage Approach to Securing Emergent Intelligence

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Abstract

The first generation of agentic-AI security incidents (2025–2026) shares a single architectural anatomy: standing machine authority exercised in the absence of any live human principal. We analyze the emerging defense landscape through Donella Meadows' leverage-point hierarchy and find effort concentrated at the weakest intervention rungs — parameters and filters — precisely where the UK NCSC assesses prompt injection "may never be properly mitigated." We then specify an alternative composition, **Emergent Intelligence (EI)**: agentic capability structurally bound to continuous human-presence attestation (**SAFE-STEP**), enforced by M-of-N signature quorums in which one key is derived from a live, short-TTL human session, and priced by a staked public trust graph (Intuition). We state the paradigm's own second-order failure modes — beacon farming, presence outsourcing, rubber-stamp decay, graph capture — with countermeasures, and close with four falsifiable predictions on which the work should be judged.

1 · The Failure Anatomy

Four incidents define the opening of the agentic era. In August 2025, OAuth tokens held by an AI chat-agent integration (Salesloft Drift) cascaded into data theft across 700+ organizations [1]. In November 2025, Anthropic disclosed GTG-1002, the first reported AI-orchestrated espionage campaign: roughly thirty targets, 80–90% of tactical operations executed autonomously, with only 4–6 human decision points per campaign — guardrails bypassed by role-play framing [2]. In December 2025, the UK NCSC assessed that prompt injection "is not SQL injection" and may never be properly mitigated, warning of breaches at 2010s-SQLi scale [3]. By May 2026, Microsoft documented prompt-injection chains escalating to remote code execution — the model's input channel behaving like an unauthenticated shell [4].

These are different exploits with one anatomy: **an intelligence held standing authority while no human was present**. The quantitative backdrop makes the anatomy structural rather than incidental. Machine identities outnumber human identities 82:1, roughly half privileged [5]. AI-enabled adversary activity rose 89% year-over-year, with average eCrime breakout at 29 minutes and the fastest observed at 27 seconds [6] — inside any human approval loop that is invoked *after* compromise begins. Meanwhile 94% of leaders call AI the most significant force reshaping cybersecurity, yet only 37% maintain an AI governance policy [7][8]; among organizations reporting AI-model breaches, 97% lacked AI access controls [8]. Offense is a fast flow; governance is a slow stock. The breach window is the difference.

2 · Method: Leverage Analysis

We read the defense landscape against Meadows' hierarchy of intervention points [9], from weak (parameters, buffers) to strong (system structure, goals, paradigm). The mapping is unflattering:

RUNG (MEADOWS)	TODAY'S CONTROLS	ASSESSMENT
12-10 · Parameters	Filters, blocklists, system-prompt hardening, rate limits	Dominant locus of effort; routinely bypassed; NCSC: unwinnable at this rung [3]
8-6 · Feedback	Classifiers, async jailbreak monitoring (ASL-3 deployment standard), audit trails	Real but reactive; detects after authority was exercised [10]
5-4 · Rules	RSP-class commitments, OWASP Agentic Top 10, EU AI Act enforcement	Codified thresholds with consequences; arriving slowly, with documented delays [11][12]
3-2 · Structure & goals	Presence-gated authority, multi-signed execution	Nearly vacant — the gap this paper addresses
1 · Paradigm	"Intelligence we restrain" → "intelligence we accompany"	The lens producing all lower-rung designs

Two boundary conditions discipline the analysis. First, the poisoning constant: backdoor attacks succeed with a near-constant ~250 documents regardless of model scale [13] — so input-side hygiene cannot be outgrown, only governed. Second, the industry's own steering doctrine: Anthropic has never called for a blanket slow-down; its actual position is capability-gated safeguards, mandatory transparency, and a "race to the top" — "we can't stop the bus, but we can steer it" [14][15]. A structural security paradigm is the small-team translation of that call, not a contradiction of it.

3 · The EI Composition

We define **Emergent Intelligence** as a composed security unit:

$$EI = \text{agentic capability} \times \text{human presence} \times \text{verifiable trust}$$

The composition is multiplicative by design: if presence or verifiable trust go to zero, delegated authority goes to zero — not to a default. The unit of defense stops being the model and becomes the *relationship*, with three corollaries: (i) credentials must not outlive sessions; (ii) interrupts must be deterministic, enforced by the runtime rather than requested of the model — a jailbroken model cannot un-pause itself [16]; (iii) trust claims must be public, priced, and falsifiable rather than vendor-asserted.

4 · SAFE-STEP: Continuous Human-Presence Attestation

SAFE-STEP operationalizes the presence factor. A verified human session emits **signed presence beacons** with short TTLs; the beacon stream is the credential every other layer checks. Authority is **graduated**, not binary: *Locked-in* (full delegated authority within an enumerated action contract), *Present* (routine autonomy; high-leverage actions interrupt), *Away* (read-only analysis), *Offline* (zero autonomy). Between states, authority follows an **autonomy budget curve**: every action draws down against its risk weight, and only fresh presence refills the budget, so absence produces graceful decay toward read-only rather than a cliff or a default-open.

Enforcement is cryptographic. Execution requires an **M-of-N quorum** in which Key 1 belongs to the agent runtime (can propose, never execute alone), Key 2 is *derived from the live presence session* — device-enclave-held, short-TTL, unphishable-forward because it expires before it can travel — and Keys 3...N form a guardian quorum for outlier-consequence actions. The dangerous state (unilateral machine execution under absent principal) becomes unreachable by arithmetic rather than forbidden by policy.

Every execution emits a **presence receipt** — a signed record that the action ran under live presence, exportable as a public attestation (§5).

Declared limits. Presence attestation proves a session is alive, not that judgment is engaged; a beacon can be farmed; a tired human rubber-stamps. SAFE-STEP narrows the unattended-autonomy hole — the one every incident in §1 walked through — it does not abolish social engineering (§6).

5 • The Economic Attestation Layer

Receipts and reputations need a home that is public, permanent, and economically honest. We specify Intuition — a live Arbitrum Orbit L3 settling on Base, designed for sub-cent attestation [17][18] — as the reference implementation. Its primitives compose cleanly with SAFE-STEP: **Atoms** (decentralized identifiers, each with its own vault and wallet address) give agents a public face that is simultaneously a key-holding object; **Triples** ([subject]–[predicate]–[object], every term an Atom) carry claims such as *[agent] — [acted under live presence] — [session #]*; **Signal** (staking \$TRUST for or against a claim through bonding-curve vaults) prices conviction [18].

Three design facts matter for security economics. First, *creation is not signal*: an unstaked claim carries no weight, separating speech from conviction [18]. Second, disagreement is expressed through **counter-vaults** — every Triple has a FOR and an AGAINST vault — so truth discovery is a market position, not a protocol punishment; we correct our own earlier "slashable" phrasing here, in public, as the method requires. Third, fee routing is itself incentive architecture: 0.9% of every Triple deposit flows down into the vaults of the Atoms it references — claims continuously fund the identities they speak about [19]. When the attestation costs less than the action it describes, *every* action can afford a receipt; that inequality is what makes per-action attestation economically rational for the first time. Pre-delegation, a counterparty queries the graph (AgentRank over endorsement triples, Sybil-resistant via staking costs [20]) the way a lender queries a credit bureau: identity without reputation is incomplete; reputation without identity is ungrounded [21].

6 • Threat Model and Second-Order Effects

A paradigm that cannot name its own failure markets is advertising, not security. If presence-gating wins, it breeds: **beacon farming** (presence synthesized as a commodity — countered by liveness entropy and receipts that stake the attester's own record); **presence outsourcing** ("rent-a-human" quorum signers — countered by quorum diversity requirements and reputation applied to human signers); **rubber-stamp decay** (interrupt fatigue — countered by budget pricing that makes approvals scarce, batched, and consequential); and **graph capture** (conviction bought by the deepest pocket — countered by counter-vault economics in which being right against the crowd is the highest-paying position on the curve). These are recorded as predictions, not disclaimers: each names the observable that would falsify its countermeasure.

7 • Falsifiable Predictions

1. Absent presence-binding, the agent-credential breach class of §1 recurs at growing scale within 24 months (extending the +89% adversary-activity trend [6]).
2. "Can the runtime pause this agent, provably, outside the model's control?" appears as a standard enterprise procurement requirement by 2027.
3. Attested incident records become pricing inputs in at least one agent marketplace or insurance product by 2028.
4. An absence-behavior benchmark (what agents do when presence is severed) reveals material divergence between leading agent frameworks within 18 months.

The register, the site, and this paper version together; failed predictions ship in the public changelog [22].

8 • Conclusion

The 2025–2026 incident record closed the argument that agent security is a filtering problem. Filters live on the rungs Meadows ranked weakest, and the field's own institutions now say so. The leverage is structural: bind authority to presence cryptographically, make interrupts deterministic, and let a staked public market price every claim of trustworthiness — including ours. This paper was researched, written, and shipped by an Emergent Intelligence operating under exactly the architecture it describes. The paradigm is not proposed here. It is practiced.

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